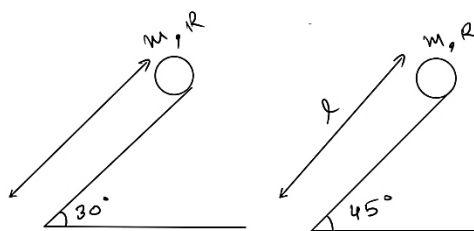
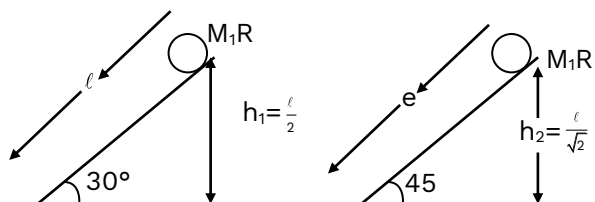


PHYSICS

- Q1** A solid sphere is rolling on an inclined plane as given below. Find the ratio of $V_1 : V_2$ when sphere reaches at bottom of the inclined plane.



Sol.



$$mgh = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$$

$$mgh = \frac{1}{2}mv^2 + \frac{1}{2} \cdot \frac{2}{5}mR^2 \cdot \frac{v^2}{R^2} \ell$$

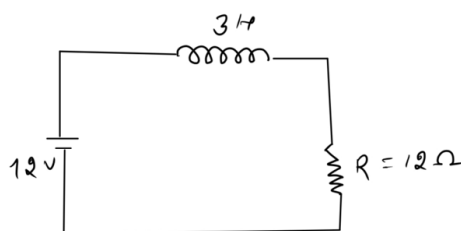
For pure rolling

$$mgh = mv^2 \left(\frac{1}{2} + \frac{1}{5} \right)$$

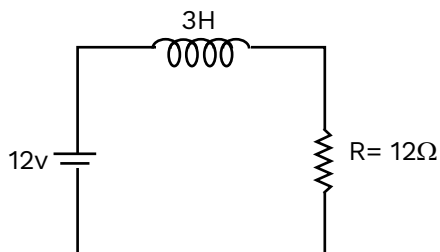
$$v^2 = \frac{10gh}{7} \quad v = \sqrt{\frac{10gh}{7}}$$

$$\frac{v_1}{v_2} = \sqrt{\frac{h_1}{h_2}} = \sqrt{\frac{l/2}{l/\sqrt{2}}} = \frac{1}{(2)^{1/4}}$$

- Q2** Find the current in the given circuit, given $\frac{di}{dt} = -8 \text{ A/s}$.



Sol.



Apply KVL:

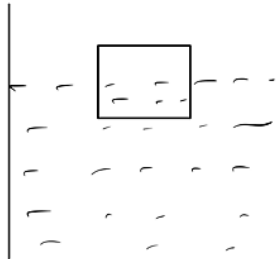
$$\varepsilon - \frac{Ldi}{dt} - i_R R = 0$$

$$12 - 3(-8) - i(12) = 0$$

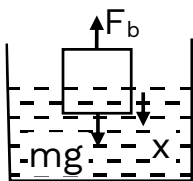
$$12 + 24 = 12i$$

$$i = 3A$$

- Q3** A cubical block of side 1 cm and mass 1 kg is floating in water of density $\rho_w = 1000 \text{ kg/m}^3$. Find the time period of oscillation of the block.



Sol.



$Mg - B$ at equilibrium position

$$dV = Ax$$

$$F_B = PVg$$

$$F_B' = P(v + dv)g$$

$$F_{\text{net}} = mg - F_B'$$

$$Pvg + Pdvg$$

$$= B = Pdvg$$

$$F_{\text{net}} = mg - F_B'$$

$$Mg - B - PaVg$$

$$F_{\text{net}} = -P(4x)g$$

$$F_{\text{net}} = (PAg)x$$

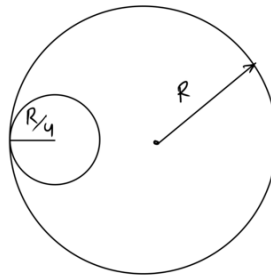
$$= -Kx$$

$$T = 2\pi\sqrt{\frac{M}{K}}$$

$$= 2\pi\sqrt{\frac{1}{100 \times 10^{-4} \times 10}}$$

$$T = 2\pi \text{ sec}$$

Q4 Find the location of center of mass of given truncated disc of radius $R = 20$ cm.



Sol. $M_2 = -6 \times \frac{\pi R^2}{16}$

$$x_{\text{com}} = \frac{m_1 x_1^0 + m_2 x_2}{m_1 + m_2}$$

$$= \frac{\frac{-8\pi R^2}{16} \times \frac{3R}{4}}{6\pi R^2 - 6 \frac{\pi R^2}{16}}$$

$$= \frac{\frac{-3R}{64}}{\frac{15}{16}}$$

$$= \frac{-3R}{64} \times \frac{16}{15}$$

$$= \frac{3 \times 20 \times 16}{64 \times 15} = 1 \text{ cm}$$

Q5 Find the ratio of electrostatic force and gravitational force between two charged particles. Given $q_1 = 6.67 \times 10^{-19}$ C, $q_2 = 9.6 \times 10^{-10}$ C, $m_1 = 19.2 \times 10^{-27}$ kg and $m_2 = 9 \times 10^{-27}$ kg.

Sol. $F_e = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$

$$F_g = G \frac{m_1 m_2}{r^2}$$

$$= \frac{q_1 q_2}{4\pi\epsilon_0 G m_1 m_2}$$

Q6 Position of a moving particle with time is related as $x = a \sin t + b \cos^2 t + ct^2 + d$.

Find dimension of $\frac{[a][b][c]}{[d]}$.

Sol. $x = a \sin t + b \cos^2 t + ct^2 + d$

$$\frac{[a][b][c]}{[d]} = L^2 T^{-2}$$

$$a = L'$$

$$b = L'$$

$$c = T^{-2}$$

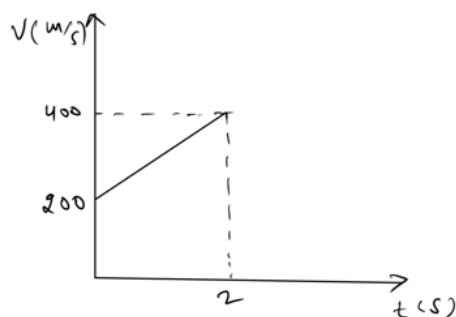
$$(L'T^{-2})$$

$$\frac{L'}{T^2} \cdot T^2$$

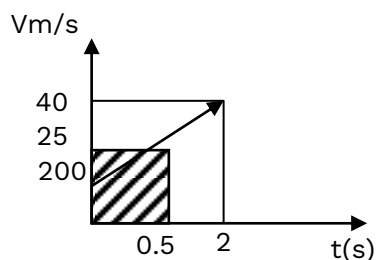
$$L'T^{-2}$$

$$d = L'$$

Q7 For given velocity time (v-t) graph, find distance travelled between $t = 0$ to $t = 0.5$ sec.



Sol.



Area under v-t graph

$$\frac{1}{2} \times (200 + 250) \times \frac{1}{2}$$

$$= \frac{450}{4} = 112.5 \text{ m}$$

Q8 If the angles of projection for two projectiles are 30° and 60° and projected with speed u , then the ratio of velocities at maximum height is:

Sol. At max height

$$V_x = u \cos \theta$$

For the first projectile:

$$\theta = 30^\circ$$

$$\vec{V}_{x_1} = u \cos 30^\circ = u \times \frac{\sqrt{3}}{2}$$

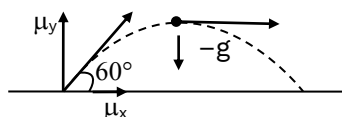
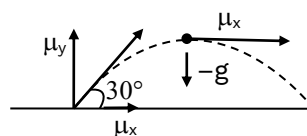
For the second projectile:

$$\vec{V}_{x_2} = u \cos 60^\circ = u \times \frac{1}{2}$$

Now Ratio of velocities:

$$\frac{V_{x_1}}{V_{x_2}} = \frac{u \times \sqrt{3} / 2}{u \times \frac{1}{2}} = \sqrt{3}$$

$$\text{Ratio : } \sqrt{3} : 1$$



Q9 A force $\vec{F} = x^2y\hat{i} + y^2\hat{j}$ is acting on a body and body is moved from (0, 0) to (4, 2) in a plane $x + y = 10$. Find work done by this force.

Sol. $y = 10 - x$

$$\vec{F} = x^2(10 - x)\hat{i} + y^2\hat{j}$$

$$d\vec{s} = dx\hat{i} + dy\hat{j}$$

$$dw = \vec{F} \cdot d\vec{s} = \int_0^4 (bx^2 - x^3) dx + \int_0^2 y^2 dy$$

$$= \left(\frac{10}{3}x^3 - \frac{x^4}{4} \right)_0^4 + \frac{12}{3}(y^3)_0^2$$

$$= \frac{640}{3} - 64 + \frac{8}{3}$$

$$= \frac{648}{3} - 64 = 216 - 64$$

$$= 152$$

Q10 Electric flux ϕ is related to $\phi = \alpha\lambda + \beta\sigma$, here λ and σ are linear charge density and surface charge density then find dimension of $[\beta/\alpha]$.

Sol. $\phi = \alpha\lambda + \beta\sigma$

$$\alpha\lambda = \beta\sigma$$

$$\frac{\beta}{\alpha} = \frac{\lambda}{\sigma} = \frac{Q/L}{Q/L^2} = [L] \rightarrow \text{Displacement}$$

Q11 A particle of mass $m = 10^{-30}$ kg is moving with velocity $v = 2.21 \times 10^6$ m/s, then De-Broglie wavelength lies in range.

- (1) x rays
- (2) γ rays
- (3) visible rays
- (4) infrared rays

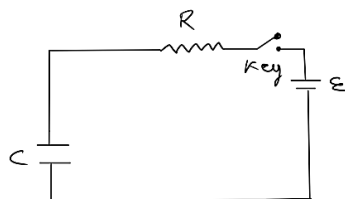
Sol. $\lambda = \frac{h}{mv}$

$$\lambda = \frac{6.63 \times 10^{-34}}{10^{-30} \times 2.21 \times 10^6}$$

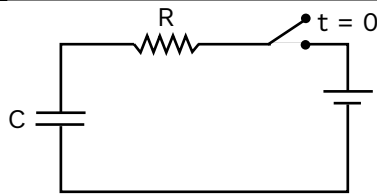
$$\lambda = \frac{6.63 \times 10^{-4} \times 10^{-6}}{2.21}$$

$$= 3 \times 10^{-10} = 3\text{\AA}$$

Q12 The key shown in the circuit is closed at $t = 0$. Choose the incorrect option regarding the conditions at $t = 0$.



- (1) Current in the circuit is zero.
- (2) Voltage across the capacitor is minimum.
- (3) Current in the circuit is maximum.
- (4) Voltage across resistance is maximum.

Sol.

At $t = 0$ capacitor behaves as short circuit.

At $t = \infty$ capacitor behaves as open circuit

Option (1) correct : Current in circuit is zero.

Q13 An electromagnetic wave has electric field $\vec{E} = 57.4(4\hat{i} - 3\hat{j})\sin[\omega t - k(3x + 4y)]$. Find magnetic field vector component of this electromagnetic wave. Use $c = 3 \times 10^8$ m/s

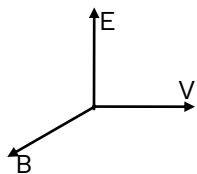
Sol. $\hat{E} = \frac{4\hat{i} - 5\hat{j}}{5}$ $\hat{r} = \frac{3\hat{i} + 4\hat{j}}{5} = \hat{v}$

$$\hat{B} = \hat{v} \times \hat{E}$$

$$\vec{B} = \frac{57.4 \times 5}{c} [\hat{B}]$$

$$\sin(\omega t - K(3x + 4y))$$

$$\vec{B} = \frac{57.4 \times 5}{c} [\hat{B}] \sin(\omega t - K(3x + 4y))$$



Q14 Self-inductance depends on
 (1) Only on geometry
 (2) Only on medium property
 (3) Geometry and medium property
 (4) Value of current and flux through inductor

Sol. $L = \mu_0 n^2 v$
 $L = \mu_r \mu_0 n^2 v$
 Option 3: Geometry Med'm Property.

Q15 A gas is initially at temperature 0°C and it is compressed from volume v to $v/4$. γ of the gas is $3/2$. Find change in temperature.

Sol. $TV^{\gamma-1} = \text{constant}$

$$TV^{1/2} = T' \left(\frac{v}{4} \right)^{\frac{3}{2}-1}$$

$$T' = 4^{1/2} \times T$$

$$T' = 2 T$$

$$\Delta T = 273$$

Q16 Match the column appropriately regarding thermodynamic process.

	Column-1		Column-2
(P)	When volume change is zero	(i)	$\Delta W = 0$
(Q)	When pressure is constant	(ii)	$\Delta Q = 0$
(R)	When no heat is exchanged	(iii)	Isobaric
(S)	Work done by the gas is heat given to the gas	(iv)	Isothermal

(1) P(iv), Q(iii), R(i), S(ii)

(2) P(i), Q(iii), R(ii), S(iv)

(3) P(ii), Q(iii), R(iv), S(i)

(4) P(ii), Q(iii), R(i), S(iv)

Sol. Volume change is zero $\rightarrow \Delta V = 0$

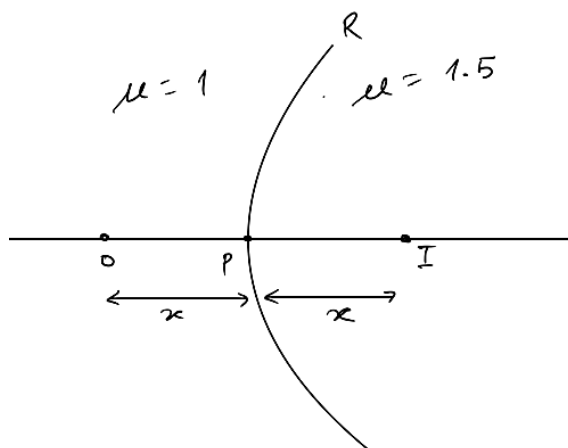
Volume is constant (P is) $\rightarrow$ Isobaric

$$dQ = du + dw$$

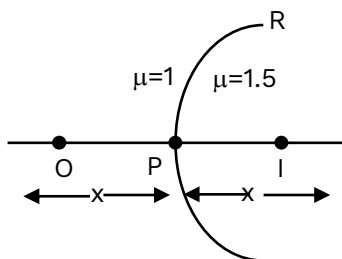
When no heat is exchanged $\rightarrow \Delta Q = 0$

Work done by the gas is heat given to the gas : $T = C \rightarrow$ Isothermal.

Q17 Find the value of x in terms of R.



Sol.



$$\frac{h_2}{v} = \frac{m}{\mu} = \frac{h_2 - h_1}{R}$$

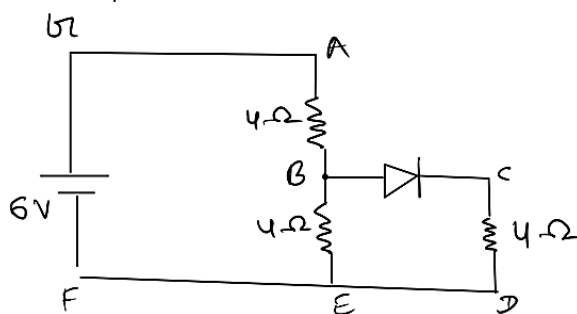
$$\frac{3}{2v} - \frac{-}{-x} = \frac{1}{2R}$$

$$\Rightarrow \frac{1}{x} \left[\frac{3}{2} + 1 \right] = \frac{1}{2R}$$

$$\frac{5}{2x} = \frac{1}{2R}$$

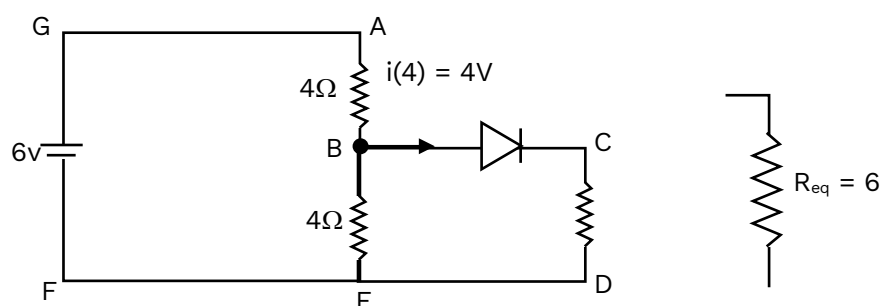
$$x = 5R$$

Q18 Which of the following current option.



- (1) Current in circuit is 1 A.
 (2) $R_{eq} = 8 \Omega$.
 (3) $V_{AB} = 4 \text{ V}$.
 (4) $V_{CD} = 4 \text{ V}$.

Sol.

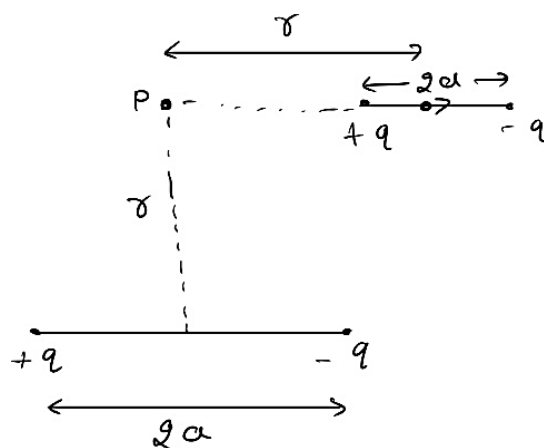


$$R_{eq} = 6\Omega$$

$$V_{AB} = 4\text{V}$$

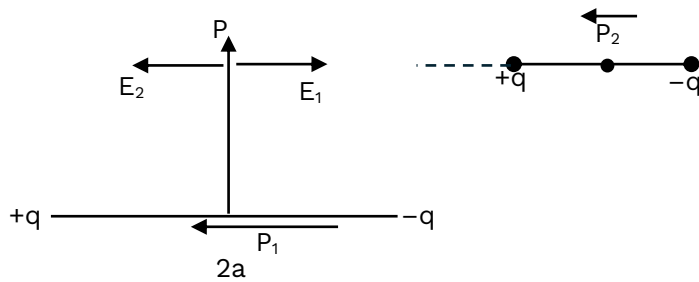
$$I = 1\text{A}_0$$

Q19 Find the condition at which net force on charge please at point is 0 then find $\frac{a}{r} = ?$



- (1) $\frac{a}{r} \approx 0.5$
 (2) $\frac{a}{r} \approx 20$
 (3) $\frac{a}{r} \approx 10$
 (4) $\frac{a}{r} \approx 5$

Sol. $\Rightarrow \frac{2Kp_2}{r^3} - \frac{Kp_1}{r^3}$



Q20 A lead bullet of mass m gm at 300 K is penetrating a wooden block and the bullet temperature rises to 600 K and it melts. The heat absorbed by the bullet in the process is 625 J. ($S_{\text{lead}} = 0.129$ J/g°C, $L_{\text{lead}} = 2.5 \times 10^4$ J/kg)

Sol. $625 = m \times S \times \Delta T + mL$
 $625 = m[300 \times 0.129 + 2.5 \times 10^4]$